An integrated scRNA-seq lung atlas built from agent-curated datasets (TODO: Be consistent with ”cell” vs ”barcode”)

Oliver Todreas[1*], Parashar Dhapola[2]

1 Deptartment of Biology/Master’s Program in Bioinformatics, Lund University, Lund, Sweden

2 Nygen Analytics AB, Lund, Sweden

* ol6437ek-s@student.lu.se

Abstract

The abundance of nucleotide sequence data has grown into the order of trillions of bases. As public archives grow in scale, so too does their fragmentation and complexity, meaning that disease-focused meta-analyses remain difficult to accomplish. We identified the need for producing atlas-scale datasets that both incorporate work spanning studies and are uniformly processed and targeted to disease areas. Here, we develop a reproducible pipeline for building a lung tissue-specific, integrated, and study-aware atlas from a broad, AI-curated atlas of single-cell RNA sequencing (scRNA-seq) data. We present an, analysis-ready atlas that successfully integrates XXX,XXX,XXX cells across X,XXX files and XXX distinct scientific projects, preserving cell-type structure and reducing batch effects.

Introduction

Public single-cell RNA sequencing (scRNA-seq) archives now contain enough cells for tissue-scale reuse, but experiments are still deposited independently, processed with different pipelines, and described with free-text metadata (Luecken et al., 2022; Mueller et al., 2026). Cross-study analysis therefore rarely starts from a shared gene axis or a study-level batch structure. Building an atlas from public data remains a separate construction problem, whereby a tissue must be selected, matrices must be brought onto a common reference, and the joint object must be integrated without erasing cell-type biology.

Community contributed collections such as CZ CELLxGENE Discover enforce cell-level and metadata standardization on submission, allowing data exploration and meta-analysis (CZI Cell Science Program et al., 2025; Rozenblatt-Rosen et al., 2017). Efforts like the Human Lung Cell Atlas integrate selected respiratory studies into a consensus cell-type map (Sikkema et al., 2023). These resources straddle the inevitable tradeoff between being exhaustive and curated. In the case of the mentioned efforts they do not ingest the Sequence Read Archive (SRA) as a whole (Leinonen et al., 2011). On the other hand, scBaseCount opts for extensiveness by using an agentic AI metadata curation approach to collect and reprocess public 10X scRNA-seq experiments at large scale, producing a library of `h5ad` files keyed by SRA experiment accession (Youngblut et al., 2025). They are neither tissue-restricted, clustered, nor treated with any batch correction.

Here we construct an integrated lung-tissue atlas from the `2026-01-12` scBaseCount human GeneFull release. We focus on lung tissue because of the frequency and severity of lung diseases. Lung cancer was estimated to be the most frequently diagnosed cancer at 2.6 million new cases, and the leading cause of cancer death with 1.9 million deaths in 2024 (Sung et al., 2026). Non-cancer diseases such as idiopathic pulmonary fibrosis are also a cause for concern and are known to be associated with high mortality rates (Chakraborty et al., 2022). scRNA-seq has promise and meaningful progess when it comes to researching lung diseases such as these (Chakraborty et al., 2022; Navin et al., 2011). To focus on lung tissue samples, we select experiments from scBaseCount whose tissue labels match lung. We apply per-dataset quality control, align datasets by gene and concatenate them over cell them into a single object. We determine that scBaseCount does not contain any cross-dataset variable that could serve as a key to correct for the batch effect. Therefore, we retrieve the National Center for Biotechnology Information's (NCBI) BioProject accessions for each dataset and use them as the Harmony batch variable (Barrett et al., 2012; Korsunsky et al., 2019). (TODO: mention that context lookup produces a json object for each SRX–here there is further information for stuff like disease state, etc) We also develop a method to optimize a clustering algorithm parameter, namely the Leiden algorithm's resolution parameter, by matching clusters to CZ CELLxGENE `cell_type` labels (Traag et al., 2019). Finally, we evaluate batch correction with a suite of metrics proposed by Luecken et al. (2022) on a study-stratified subset. The result is an analysis-ready lung atlas derived from an agent-curated public resource.

Materials and methods

Building an integrated lung atlas using scBaseCount data required developing robust single-cell workflows in Python using Scanpy (Wolf et al., 2018). We enforced observability through extensive logging configured with the Python logger.

Data aquisition

scBaseCount data, which includes expression matrices, are hosted on Google Cloud Platform in Google Cloud Storage (GCS) buckets (Youngblut et al., 2025). We used human, GeneFull matrices from the `2026-01-12` release of scBaseCount. Before downloading any matrices, we identified a subset of the available matrices from corresponding metadata table shipped with scBaseCount. We discarded accessions that were not SRA or ENA records (identifiers beginning with NRX). We then flagged lung tissue by applying a regular expression to the free-text tissue field, matching terms such as "lung," "pulmonary," "alveolus," and "bronchus." For each remaining accession we retrieved a BioProject accession from the European Nucleotide Archive (ENA) portal API (Amid et al., 2020). We excluded seven BioProjects known to mix lung with many other organs (PRJEB51634, PRJNA1005589, PRJNA1179423, PRJNA1188170, PRJNA1215450, PRJNA657844, and PRJNA902813), dropped accessions that lacked a BioProject accession, and dropped studies whose combined cell count was 1,000 or fewer. We sequentially fetched datasets that remained after these filters from GCS and computed their MD5 hashes on disk. Due to the extensive cumulative size of the datasets, we uploaded the datasets themselves, along with the newly computed hashes, to an internally owned and maintained Cloudflare R2 bucket, serving as a production mirror of scBaseCount, allowing us to remove local files from disk.

Atlas construction and quality control

To integrate the aquired individual datasets as a single atlas-scale `h5ad` file, we built a pipeline to sequentially download scBaseCount datasets from the R2 mirror, filter them by performing quality control operations, determine if what remained of the filtered datasets was sufficient for inclusion in the atlas, align the datasets to a fixed gene axis, and concatenate the datasets to one in-memory atlas. We wrote the concatenated atlas to disk on pipeline completion.

To ensure datasests matched scBaseCount, we computed dataset MD5 hashes following download from R2 and compared them the MD5 hashes stored as metadata alongside the corresponding R2 objects, the latter having been computed upon upload. We enforced identity between the upload and download hashes.

Next, we filtered cells based on tunable parameters. We did not drop genes for rarity. We flagged genes as likely mitochondrial, ribosomal, and hemoglobin genes by applying regular expressions to the gene names, and computed per-cell quality metrics. We first dropped cells with a total gene count below 200 genes. We dropped cells whose mitochondrial and ribosomal gene counts were at least 20% and 50%, respectively, of total gene counts. We did not filter cells on hemoglobin gene counts. Following these filters, we applied a cutoff on number of remaining cells and percentage of cells remaining relative to the initial number of cells in the dataset. We did not include datasets in the atlas if they had fewer than 100 cells remaining or if fewer than 50% of cells remained in the dataset following filtering. We also excluded datasets in which every cell lacked a usable CZ CELLxGENE `cell_type` annotation (CZI Cell Science Program et al., 2025). We configured every mentioned threshold as a tunable parameter. To ensure that the values we settled on were well calibrated, we recorded the number of cells dropped by each filter, allowing us to rerun the filtering pipeline with different filtering parameters. We did not apply doublet detection. We chose filtering parameters based on recommendataions proposed by Luecken and Theis (2019) and enriched them with the ribosomal filter and hemoglobin flagging. Before we concatenated the filtered datasets along the cell axis to the atlas `h5ad` file, we reindexed each dataset onto the scBaseCount `geneInfo.tab` gene list, discarding genes absent from that list and filling missing reference genes with zeros, to ensure concatenated matrices used a shared gene axis. Since Youngblut et al. (2025) already indexed scBaseCount matrices against a shared STAR reference, this reindex did not change the genome reference.

Embedding and batch correction

We developed a method to embed and cluster the atlas using the CZ CELLxGENE `cell_type` key present in the datasets. First, we normalized gene counts per cell so that each cell had the same total gene count. Then, we logarithmized the data matrix by the natural logarithm, selected the top 2,000 highly variable genes (HVG) stratified by BioSample—a number in the range of recommended HVGs according to Luecken and Theis (2019)—and z-scaled the expression of each gene. We then applied principal component analysis (PCA), computing the first 50 PCs. We then recorded the minimum number of PCs in the range of 15–50 explained at least 50% of the data's cumulative variance. We applied the processing steps through PCA using Scanpy (Wolf et al., 2018).

Following PCA, we applied Harmony to the PCs to correct for one batch effect (Korsunsky et al., 2019). The scBaseCount release from which we collected data did not contain any keys coarser than SRA experiment accession, which was effectively a one-to-one file key. We identified the NCBI BioProject accession as a key to use for applying batch correction to the atlas. BioProject accessions group SRA experiment accessions by project type, project scope, technical methodology, or other groupings (Barrett et al., 2012). To collect the BioProject accession for a given SRA experiment accession, along with other relevant metadata, we queried the European Nucleotide Archive's (ENA) portal API (Amid et al., 2020). As a result of the International Nucleotide Sequence Database Collaboration (INSDC), ENA and NCBI both host BioProject accessions (Arita et al., 2021).

To vizualise and cluster the data, we computed the nearest neighbors distance matrix and a neighborhood graph of observations and applied Scanpy's reproducible, single threaded UMAP dimensionality reduction implementation (McInnes et al., 2018). We clustered the neighborhood graph using the Leiden algorithm (Traag et al., 2019). To select an optimal Leiden clustering resolution, we sequentially applied Leiden clustering to the neighborhood graph at a range of resolutions from 0.1–2.0 at intervals of 0.1, computed the Jaccard index (also known as "intersection over union" or "IoU") of each CZ CELLxGENE `cell_type`-Leiden cluster pair, and determined the optimal matching using SciPy's linear sum assignment implementation (Virtanen et al., 2020). We recorded the cost of the assignment at each Leiden clustering resolution, and retained the Leiden clustering resolution with the lowest cost as the cannonical Leiden clustering resolution. TODO: decide on if/how to include RF.

To test our clustering methodology, we implemented it as a set of importable functions and applied them to datasets corresponding with single SRA experiment accessions before applying it to the atlas. We took a stratified sample of 100,000 cells, where strata are BioProjects and counts per stratum are proportional to each BioProject's share of the full atlas (with at least one cell per BioProject). We benchmarked the batch key using scIB (Luecken et al., 2022).

Results and discussion

There were 35,266 datasets available in the `2026-01-12` release of scBaseCount. We discarded X,XXX accessions whose identifiers were not SRA or ENA records. We retained accessions whose free-text tissue label matched a lung-related regular expression, leaving X,XXX lung-tissue accessions. After retrieving study accessions from the ENA portal API, we dropped X,XXX of those accessions from seven multi-organ studies, and any accession that lacked a study accession. We then dropped studies whose combined cell count was 1,000 or fewer, removing X,XXX further accessions. The remaining X,XXX accessions (X,XXX,XXX cells; mean X,XXX, median X,XXX cells per accession) formed the eligible set.

The atlas contained X,XXX datasets from X,XXX eligible datasets, spanning XXX BioProject accessions. After filtering, the atlas contained X,XXX,XXX barcodes, down from X,XXX,XXX barcodes (Table 1). The atlas contained XX CZ CELLxGENE `cell_type` labels (Fig 1).

Table 1. Atlas contained X,XXX datasets (X,XXX,XXX barcodes) across XXX BioProject accessions, of X,XXX eligible datasets (X,XXX,XXX barcodes) following filtering.

Filter	Datasets dropped, n (%)	Barcodes dropped, n (%)
MD5 hash (mismatch)	0 (0)	X,XXX (X)
CZ CELLxGENE `cell_type` (labels absent)	XXX (X)	XX,XXX (X)
Min. genes per cell ($<$ 200)	—	X,XXX (X)
Max. mitochondrial fraction ($\geq$ 20%)	—	XXX (X)
Max. ribosomal fraction ($\geq$ 50%)	—	XX (X)
Min. cells remaining ($<$ 100)	XX (X)	XX,XXX (X)
Min. fraction of cells remaining ($<$ 50%)	XX (X)	XX,XXX (X)

A total of the XXX datasets were dropped. Of those, 0 were dropped due to MD5 hash mismatch, XXX (X% total datasets) were dropped due to fully absent CZ CELLxGENE `cell_type` labels. Of the XX,XXX barcodes that were dropped, X,XXX (X% total barcodes) were dropped for gene counts of less than 200, XXX (X% total barcodes) were dropped for having 20% or more of total gene count being mitochondrial, and XX (X% total barcodes) for having 50% or more of total gene counts being ribosomal. Following filtering, datasets dropped for excessive barcode dropout. XX (X% total datasets) were dropped for retaining fewer than 100 cells, and XX (X% total datasets) were dropped for retaining less than 50% of cells following filtering.

On atlas scale, batch effect removal per CZ CELLxGENE `cell_type` label, measured by the k-nearest-neighbor batch effect test (kBET) proposed by Büttner et al. (2019) and implemented by Luecken et al. (2022) was lower from the uncorrected PCs compared to the batch corrected PCs, the latter using NCBI BioSample as batch key (TODO:decide whether to cite all scib metrics in text). The aggregate biological conservation score proposed by Luecken et al. (2022) decreased from 0.53 to 0.51 as a result of batch correction, while the batch correction aggregate score increased from 0.28 to 0.45 (Fig 2).

It is clear that bat

We validated our resolution selection methodology on five lung experiment files from the same release, sampled at the 25th, 33rd, 50th, 67th, and 75th percentiles of cell count (X,XXX–X,XXX cells before filtering). On each file we ran Leiden clustering at resolutions 0.1–2.0 in steps of 0.1, scored each partition by the summed Jaccard index of the optimal one-to-one matching between clusters and CZ CELLxGENE `cell_type` labels, and retained the resolution with the highest score. Selected resolutions ranged from X.X to X.X, and cluster counts were close to the number of retained cell-type labels. Silhouette scores generally

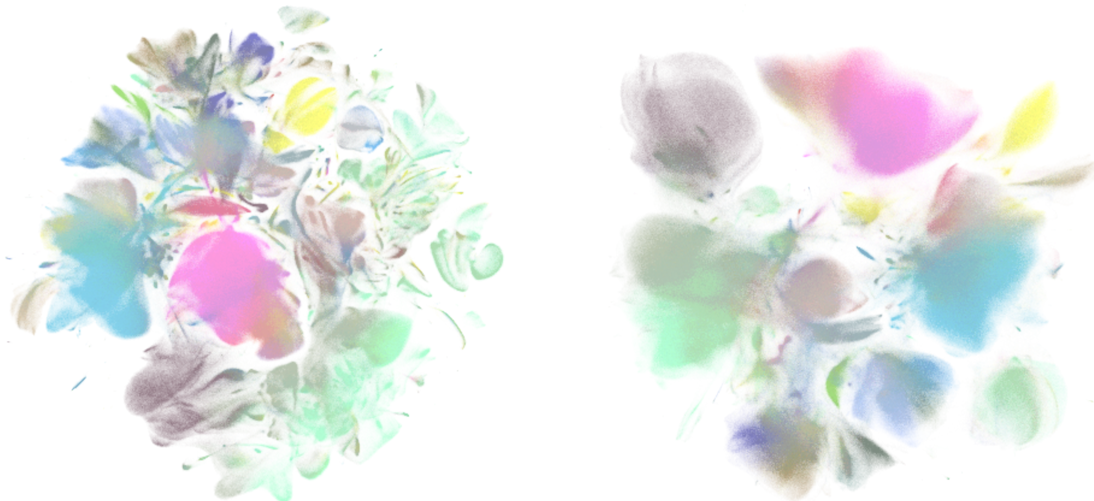

Fig 1. Single-file integrated lung disease atlas derived from scBaseCount. UMAP of the integrated atlas colored by CZ CELLxGENE `cell_type`. A: Uncorrected embedding. B: Harmony-corrected embedding (batch key: BioProject).

	Bio conservation					Batch correction					Aggregate score		
Method	Isolated labels	KMeans NMI	KMeans ARI	Silhouette label	cLISI	BRAS	iLISI	KBET	Graph connectivity	PCR comparison	Batch correction	Bio conservation	Total
X_pca_harmony	0.45	0.46	0.20	0.44	0.99	0.75	0.02	0.48	0.47	0.52	0.45	0.51	0.49
X_pca	0.50	0.51	0.21	0.45	0.99	0.50	0.00	0.31	0.60	0.00	0.28	0.53	0.43

Fig 2. Batch correction metrics before and after Harmonization of PCs. Benchmark results on a stratified atlas subset comparing uncorrected PCs and Harmony-corrected PCs (batch key: NCBI BioSample). Aggregate biological conservation decreased from 0.53 to 0.51; aggregate batch correction increased from 0.28 to 0.45. kBET was lower for uncorrected than batch-corrected embeddings.

fell as resolution increased, and the Jaccard-selected resolution was never the silhouette maximum: silhouette rewards compact, well-separated groups, whereas the cell-type prior often sits inside larger partitions. On a 100,000-cell, study-stratified subset of the atlas the same matching criterion favoured resolution X.X (XX clusters in the subset). We used that resolution for the Harmony-corrected atlas, which produced XX clusters.

Continued study is warranted using the present atlas. Due to the atlas spanning over 100 BioProject accessions, we recognize that differential gene expression analyses on various clusters has the potential to reaffirm disease state gene expression in certain cell types, or to highlight novel, low signal relationships that have the potential to represent new findings. As evident in the UMAP projection of the atlas, certain CZ CELLxGENE `cell_type` labels tend to cluster naturally, such as B and T lymphocytes (Fig 1 B). Sant et al.

(2025) note that a cluster's ability to reliably representat real cellular identity and state is crucial for deriving biological meaning in downstream analysis steps. Since the clustering of the present atlas builds on the CZ CELLxGENE `cell_type` annotations as priors informing the clustering resolution, the clusters may indeed represent real biology.

Furthermore, we identified limitations with the study. Starting with atlas construction, reproduction requires cloud storage in the order of hundreds of GB. The concatenation of `h5ad` files was optimized for limited disk space rather than limited memory, and therefore requires that the entire concatenated atlas is held in memory before it is written to disk. Realistically, memory in the order of TB is required to safely reproduce the atlas or build a new one with the code present to avoid out-of-memory errors late in the concatenation. Additionally, given that scBaseCount is agentically curated, it is not surprising that we identified certain `h5ad` files that upon close inspection using the ENA portal API did not appear to entirely match the metadata perscribed by the agent built by Youngblut et al. (2025). While we sought out to identify reliably accurate fields in our deterministic context lookup on ENA via SRA experiment accession, it is entirely possible that certain metadata fields we rely on for context lookup can be missing or inaccurate, which can negatively affect downstream analysis, which relies on assumptions about which cells are in which state.

Conclusion

The field of single-cell technology continues to grow rapidly. With it, the availability of data, tools, and analysis methodology, has become difficult to navigate (Heumos et al., 2023). Meanwhile, expanded cell counts in scRNA-seq datasets is crucial for executing analysis of complex cell populations with rare cell types and states (Kolodziejczyk et al., 2015). Nonetheless, simply increasing the number of cells considered in scRNA-seq data analysis is not sufficient for generating those large datasets (Mueller et al., 2026). To produce processed datasets at large scale represents a challenge spanning data aquisition, processing, and correction for batch effects that materialize at atlas-level. Here, we propose a method for identifying and utilizing publically available resources in the form of large raw matrix stores—in this case scBaseCount—and integrating data to produce a single, analysis-ready atlas with extensive processing applied (Youngblut et al., 2025).

The atlas building pipeline proposed here does not exist in isolation. During the writing of the present report, a Snakemake workflow for concatenating raw count matrices, and applying quality control and batch correction was developed by Mueller et al. (2026), a method that addresses the growing demand for curated atlases to be used for downlstream data analysis. Continued work would certainly be warranted in implementing the pipeline built by Mueller et al., 2026, as a means of validating the quality control and integration decisions made in the present method or producing more atlases.

References

Amid, C., Alako, B. T. F., Balavenkataraman Kadhirvelu, V., Burdett, T., Burgin, J., Fan, J., Harrison, P. W., Holt, S., Hussein, A., Ivanov, E., Jayathilaka, S., Kay, S., Keane, T., Leinonen, R., Liu, X., Martinez-Villacorta, J., Milano, A., Pakseresht, A., Rahman, N., … Cochrane, G. (2020). The European Nucleotide Archive in 2019. *Nucleic Acids Research*, *48*(D1), D70–D76. https://doi.org/10.1093/nar/gkz1063

Arita, M., Karsch-Mizrachi, I., Cochrane, G., & on behalf of the International Nucleotide Sequence Database Collaboration. (2021). The international nucleotide sequence database collaboration. *Nucleic Acids Research*, *49*(D1), D121–D124. https://doi.org/10.1093/nar/gkaa967

Barrett, T., Clark, K., Gevorgyan, R., Gorelenkov, V., Gribov, E., Karsch-Mizrachi, I., Kimelman, M., Pruitt, K. D., Resenchuk, S., Tatusova, T., Yaschenko, E., & Ostell, J. (2012). BioProject and BioSample databases at NCBI: Facilitating capture and organization of metadata. *Nucleic Acids Research*, *40*(D1), D57–D63. https://doi.org/10.1093/nar/gkr1163

Büttner, M., Miao, Z., Wolf, F. A., Teichmann, S. A., & Theis, F. J. (2019). A test metric for assessing single-cell RNA-seq batch correction. *Nature Methods*, *16*(1), 43–49. https://doi.org/10.1038/s41592-018-0254-1

Chakraborty, A., Mastalerz, M., Ansari, M., Schiller, H. B., & Staab-Weijnitz, C. A. (2022). Emerging Roles of Airway Epithelial Cells in Idiopathic Pulmonary Fibrosis. *Cells*, *11*(6), 1050. https://doi.org/10.3390/cells11061050

CZI Cell Science Program, Abdulla, S., Aevermann, B., Assis, P., Badajoz, S., Bell, S. M., Bezzi, E., Cakir, B., Chaffer, J., Chambers, S., Cherry, J. M., Chi, T., Chien, J., Dorman, L., Garcia-Nieto, P., Gloria, N., Hastie, M., Hegeman, D., Hilton, J., … Carr, A. (2025). CZ CELLxGENE Discover: A single-cell data platform for scalable exploration, analysis and modeling of aggregated data. *Nucleic Acids Research*, *53*(D1), D886–D900. https://doi.org/10.1093/nar/gkae1142

Heumos, L., Schaar, A. C., Lance, C., Litinetskaya, A., Drost, F., Zappia, L., Lücken, M. D., Strobl, D. C., Henao, J., Curion, F., Schiller, H. B., & Theis, F. J. (2023). Best practices for single-cell analysis across modalities. *Nature Reviews Genetics*, *24*(8), 550–572. https://doi.org/10.1038/s41576-023-00586-w

Kolodziejczyk, A. A., Kim, J. K., Svensson, V., Marioni, J. C., & Teichmann, S. A. (2015). The Technology and Biology of Single-Cell RNA Sequencing. *Molecular Cell*, *58*(4), 610–620. https://doi.org/10.1016/j.molcel.2015.04.005

Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P.-r., & Raychaudhuri, S. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, *16*(12), 1289–1296. https://doi.org/10.1038/s41592-019-0619-0

Leinonen, R., Sugawara, H., Shumway, M., & on behalf of the International Nucleotide Sequence Database Collaboration. (2011). The Sequence Read Archive. *Nucleic Acids Research*, *39*(suppl_1), D19–D21. https://doi.org/10.1093/nar/gkq1019

Luecken, M. D., Büttner, M., Chaichoompu, K., Danese, A., Interlandi, M., Mueller, M. F., Strobl, D. C., Zappia, L., Dugas, M., Colomé-Tatché, M., & Theis, F. J. (2022). Benchmarking atlas-level data integration in single-cell genomics. *Nature Methods*, *19*(1), 41–50. https://doi.org/10.1038/s41592-021-01336-8

Luecken, M. D., & Theis, F. J. (2019). Current best practices in single-cell RNA-seq analysis: A tutorial. *Molecular Systems Biology*, *15*(6), MSB188746. https://doi.org/10.15252/msb.20188746

McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). UMAP: Uniform Manifold Approximation and Projection. *Journal of Open Source Software*, *3*(29), 861. https://doi.org/10.21105/joss.00861

Mueller, M. F., Cujba, A.-M., Romanovskaia, D., Cohen, C. J., Bright, C. A., Lance, C., Ramirez-Suastegui, C., Strobl, D. C., Yuan, H., Hulsen, J., Naas, J., Limbeck, K., Kock, K. H., Halle, L., Knoll, R., Kfuri-Rubens, R., Aguilar-Fernandez, S., Parikh, S., Shitov, V. A., … Luecken, M. D. (2026, August). Building optimized single-cell reference atlases with scAtlasTb. https://doi.org/10.64898/2026.07.30.741695

Navin, N., Kendall, J., Troge, J., Andrews, P., Rodgers, L., McIndoo, J., Cook, K., Stepansky, A., Levy, D., Esposito, D., Muthuswamy, L., Krasnitz, A., McCombie, W. R., Hicks, J., & Wigler, M. (2011). Tumour evolution inferred by single-cell sequencing. *Nature*, *472*(7341), 90–94. https://doi.org/10.1038/nature09807

Rozenblatt-Rosen, O., Stubbington, M. J. T., Regev, A., & Teichmann, S. A. (2017). The Human Cell Atlas: From vision to reality. *Nature*, *550*(7677), 451–453. https://doi.org/10.1038/550451a

Sant, C., Mucke, L., & Corces, M. R. (2025). CHOIR improves significance-based detection of cell types and states from single-cell data. *Nature Genetics*, *57*(5), 1309–1319. https://doi.org/10.1038/s41588-025-02148-8

Sikkema, L., Ramírez-Suástegui, C., Strobl, D. C., Gillett, T. E., Zappia, L., Madissoon, E., Markov, N. S., Zaragosi, L.-E., Ji, Y., Ansari, M., Arguel, M.-J., Apperloo, L., Banchero, M., Bécavin, C., Berg, M., Chichelnitskiy, E., Chung, M.-i., Collin, A., Gay, A. C. A., … Theis, F. J. (2023). An integrated cell atlas of the lung in health and disease. *Nature Medicine*, *29*(6), 1563–1577. https://doi.org/10.1038/s41591-023-02327-2

Sung, H., Filho, A. M., Laversanne, M., Ferlay, J., Siegel, R. L., Soerjomataram, I., Jemal, A., & Bray, F. (2026). Global cancer statistics 2024: GLOBOCAN estimates of incidence and mortality worldwide for 34 cancers in 186 countries. *Ca*, *76*(4), e70090. https://doi.org/10.3322/caac.70090

Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, *9*(1), 5233. https://doi.org/10.1038/s41598-019-41695-z

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … van Mulbregt, P. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*(3), 261–272. https://doi.org/10.1038/s41592-019-0686-2

Wolf, F. A., Angerer, P., & Theis, F. J. (2018). SCANPY: Large-scale single-cell gene expression data analysis. *Genome Biology*, *19*(1), 15. https://doi.org/10.1186/s13059-017-1382-0

Youngblut, N. D., Carpenter, C., Prashar, J., Ricci-Tam, C., Ilango, R., Teyssier, N., Konermann, S., Hsu, P. D., Dobin, A., Burke, D. P., Goodarzi, H., & Roohani, Y. H. (2025, May). scBaseCount: An AI agent-curated, uniformly processed, and continually expanding single cell data repository. https://doi.org/10.1101/2025.02.27.640494